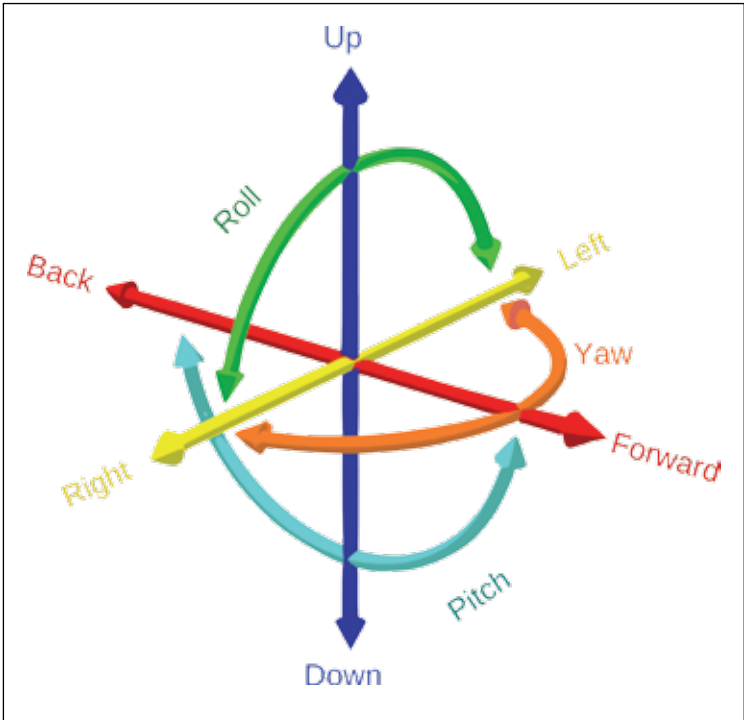




This diagram shows the degrees of freedom with respect to the 3-axes

accelerometers, gyroscopes and magnetometers detect velocity, orientation and gravitational forces. Earlier, these devices were only used in airplane orientation, but over the years cost has been struck down enough for them to be used in smartphones. Only one catch – these sensors can only detect orientation values, not translation. Hence they are only able to accomplish the rotational DoF, not the lateral ones. On that note, one more thing needs to be clarified, there is nothing called 9DoF which some VR products are claiming to have achieved. It is an incorrect summation of 3 sensors having 3DoF each.

For the smartphone-fitted headsets, the smartphone itself contains all the sensors required for the tracking, whereas the externally-powered headsets have sensors built into them that send the data to the external computational system which in turn sends back appropriate visuals. The problem of Positional Tracking, that is detecting lateral movements, is